Benders Decomposition of a Stochastic Problem

UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

Stochastic Programming

Assignment

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Abstract

This report presents the implementation of Benders decomposition for a two-stage stochastic optimization problem involving ATM cash management. The objective is to determine the optimal amount of money to deposit in an ATM before the weekend while minimizing both holding costs and expected shortage penalties under uncertain demand scenarios.

The stochastic problem is first formulated in its extensive form and then decomposed into a master problem and a recourse subproblem using Benders decomposition. The complete implementation has been developed entirely in AMPL through the use of `.mod`, `.dat`, and `.run` files, without relying on external programming interfaces.

The computational experiments show that the Benders decomposition algorithm converges in only five iterations and produces the same optimal solution as the extensive form formulation, validating the correctness of the implementation.

1 Introduction

Stochastic optimization problems arise naturally in decision-making situations where part of the information is uncertain at the moment decisions must be taken. In this assignment, we study a stochastic ATM refill problem in which a bank branch must decide how much money to deposit in an ATM before the weekend, while future cash demand is uncertain.

The objective is to minimize the total expected cost, which includes:

- a holding cost associated with the deposited cash,
- and a penalty cost when the deposited amount is insufficient to satisfy demand.

The problem can be formulated as a two-stage stochastic linear program. The first-stage decision corresponds to the amount of money initially deposited in the ATM, while the second-stage recourse decisions correspond to the shortage amounts for each demand scenario.

This report develops:

- the extensive form formulation of the stochastic problem,
- the derivation of the Benders decomposition approach,
- the implementation in AMPL,
- and the computational validation of the obtained solution.

2 Stochastic Optimization Problem Formulation

The ATM refill problem is defined using the following parameters:

- c: holding cost per unit deposited in the ATM,
- q: shortage penalty cost per unit,
- l: minimum ATM capacity,
- u: maximum ATM capacity,
- ξ_i: demand realization for scenario i,
- p_i: probability associated with scenario i,
- s: number of scenarios.

The decision variable is:

- x: amount of money deposited before the weekend.

Additionally, for every scenario i, we define:

- y_i: shortage amount when demand exceeds the deposited quantity.

The stochastic optimization problem in extensive form is:

$$\min \; cx + \sum_{i=1}^{s} p_i q y_i$$

subject to

$$l \leq x \leq u$$

$$x + y_i \geq \xi_i \qquad i = 1, \ldots, s$$

$$y_i \geq 0 \qquad i = 1, \dots, s$$

This formulation minimizes the sum of:

- the deterministic holding cost,
- and the expected shortage penalty over all demand scenarios.

3 Benders Decomposition

The problem can be naturally decomposed into:

- a master problem containing the first-stage decision variable x,
- and a subproblem evaluating the recourse cost for fixed values of x.

3.1 Master Problem

A new variable θ is introduced to approximate the expected second-stage cost.

The master problem is formulated as:

$$\min \ cx + \theta$$

subject to

$$l \leq x \leq u$$

together with the Benders cuts generated iteratively.

Initially, no cuts are present, so the master problem only optimizes the first-stage cost.

3.2 Subproblem

For a fixed value $x = \bar{x}$, the recourse problem becomes:

$$Q(\bar{x}) = \min \sum_{i=1}^{s} p_i q y_i$$

subject to

$$\bar{x} + y_i \geq \xi_i \qquad i = 1, \dots, s$$

$$y_i \geq 0 \qquad i = 1, \dots, s$$

Since the subproblem is always feasible for any x, only optimality cuts are required.

The dual formulation of the subproblem is:

$$\max \sum_{i=1}^{s} u_i(\xi_i - \bar{x})$$

subject to

$$0 \leq u_i \leq p_i q \qquad i = 1, \ldots, s$$

The dual variables u_i are then used to generate Benders optimality cuts.

3.3 Optimality Cuts

At iteration k, after solving the subproblem for the current value $\bar{x}$, the following cut is added to the master problem:

$$\theta \geq \sum_{i=1}^{s} u_i \xi_i - \left(\sum_{i=1}^{s} u_i \right) x$$

These cuts progressively improve the approximation of the expected recourse function until convergence is achieved.

4 AMPL Implementation

The implementation has been developed entirely in AMPL using three files:

- `atm_benders.mod`
- `atm_benders.dat`
- `atm_benders.run`

4.1 Model File

The `.mod` file contains:

- the extensive form formulation,
- the master problem,
- the subproblem,
- and the dynamic cut structure.

The main parameters are:

```
param c;
param q;
param l;
param u;
param s integer;
```

The scenario data are stored through:

```
set SCENARIOS := 1..s;

param p{SCENARIOS};
param xi{SCENARIOS};
```

The first-stage variable is:

```
var x >= l <= u;
```

while the second-stage variables are:

```
var y{SCENARIOS} >= 0;
```

The Benders cuts are incorporated dynamically using indexed parameters and constraints.

4.2 Run File

The `.run` file implements the iterative Benders decomposition algorithm directly in AMPL.

The algorithm proceeds as follows:

1. Solve the extensive form for validation.
2. Initialize the master problem without cuts.
3. Solve the master problem.
4. Fix the value of x.
5. Solve the subproblem.
6. Generate a new optimality cut.
7. Update upper and lower bounds.
8. Repeat until convergence.

The stopping criterion is based on the optimality gap:

$$|UB - LB| \leq \varepsilon$$

with tolerance:

$$\varepsilon = 10^{-8}$$

4.3 Data File

The `.dat` file contains the parameters provided in the assignment statement.

The problem data are:

Scenario	Probability p_i	Demand ξ_i
1	0.04	150
2	0.09	120
3	0.10	110
4	0.21	100
5	0.27	80
6	0.23	60
7	0.06	50

with:

$$c = 0.00025$$
$$q = 0.0011$$
$$l = 21$$
$$u = 147$$

5 Computational Results

The computational experiments were performed using AMPL together with the CPLEX solver.

5.1 Extensive Form Solution

The extensive form formulation produced the following optimal solution:

$$x^* = 110$$

with objective value:

$$z^* = 0.03025$$

The corresponding shortage values are:

Scenario	y_i
1	40
2	10
3	0
4	0
5	0
6	0
7	0

5.2 Benders Decomposition Results

The Benders decomposition algorithm converged after 5 iterations.

The final solution obtained was:

$$x^* = 110$$

with objective value:

$$z^* = 0.03025$$

The generated cuts progressively tightened the lower bound until convergence was achieved.

The iteration process is summarized below:

Iteration	x	Lower Bound	Upper Bound
1	21.0000	0.00525	0.07807
2	87.2000	0.02180	0.03328
3	110.9091	0.02773	0.03035
4	103.2258	0.02953	0.03027
5	110.0000	0.03025	0.03025

The final optimality gap was essentially zero.

6 Validation and Conclusions

The results obtained from the Benders decomposition algorithm exactly match those obtained from the extensive form formulation, validating the correctness of the implementation.

The decomposition converged rapidly, requiring only five iterations to reach optimality. This confirms the effectiveness of Benders decomposition for two-stage stochastic linear programming problems.

Although the current problem instance is relatively small and can easily be solved directly through the extensive form, Benders decomposition becomes especially useful for large-scale stochastic optimization problems where the extensive formulation may become computationally expensive.

The implementation also demonstrates how AMPL can be used not only for mathematical modeling, but also for controlling iterative decomposition algorithms directly through `.run` scripts without requiring external programming languages.